

ActInf Textbook Group ~ Cohort 4 ~ Meeting 20 (Chapter 9 part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 20 (Chapter 9 part 2) | 56:48

all right hello everyone welcome back
it's our second discussion on chapter n
so we can look at the
chapter and look at previous questions
but before we do so does anyone just
want to give any general comment or
thought or question on
nine
like any section of nine that was
interesting or
any
topic sorry I'm running a bit behind in
the reading so uh haven't looked at
chapter
n or or even
another
topic or another question if you want to
go to
it that's sorry
yeah
go for itna oh no I just I'm this is my
first time attending and I did not know
that we were going to be working on
chapter nine so now that I I'll be I'll
prepare but I was focusing uh I focus my
work mainly on making sure that I
understand some key
Concepts in the book and so so I'm not
really quite
ready
today oh it's all good thank you for
joining I
mean it's always listed here but also
it's kind of like we even talk about
things outside the textbook so we can

talk about whatever people
want even just the theme of chapter nine
with um modelbased data analysis I mean
this is the chapter where the models are
connected to empirical data that's
always a useful topic
or or just anything people want to talk
about or we can look at some prior
questions
I the only
thing just a really minor comment about
the book uh I just noticed that Parts
one and two are not don't have
labels on the index
page that's that's uh that's one thing
that I picked out but um but like I said
I'm I'm new to this topic I will get I
will do a lot of reading this week but I
really wanted to make sure that I
understood where the free energy
principle concept came from so I'm
really way
behind so it's all good it's all good um
when you say part one and part two you
mean chapters 1 through five and chapter
6 through 10 uh well when I have the PDF
and on the PDF on the first on the index
page I not is that that Parts one and
two there no there's no label it just
has the list of the
sections like
this let me see part one and part two
part one and part
two
y on the on right on the in the content
page
yeah so the label the the label of the
labels of the parts are not in right on

included and I didn't know if that was deliberate

or yeah I think the part one and part two is just kind of like an informal okay like the chapters are the main components

yeah um and if you want to know like for the origin the fvp chapter 10 might give you some

information oh good okay it's yeah okay any other chapter 9 related topics um or Andrew had some things to discuss or anyone else could bring something

up yeah or go for it yes please

oh okay thanks Daniel thanks Angel um

yes I have a couple questions regarding chapter uh

nine

um so

um I wonder how should I uh share with please add them to the questions

everyone is just overwhelmingly encouraged to add the written questions to the

table

write questions uh in the table yeah

where make a blank

row and just put it into the question

part okay

okay um yeah let me um okay so I have to

go to this uh this website right to

share the

questions yeah you can follow like the

the you can click like the token for

active inference to follow to where I am

in chapter

nine yes I find chapter nine yeah so add

add the questions

there

okay okay then let me check if I'm on

the same page as you do yeah uh chapter

nine yeah yeah yeah I'm I'm there so so

okay okay so I just juggle between U so

first um I have a question regarding uh

regarding

equation

9.2 um so so what

is um this

Theta

Alpha in

equation 9.2 and similar writing

did you enter the question or where did

you write okay okay what is I'm gonna I

just copied it into there but yeah oh at

the wrong place okay no no it's okay

it's okay just this is how we can get

all the questions together what is Theta

Alpha so because it hasn't show up in

our previous you know chapters before

right this Theta α sub

Alpha yes Theta sub Alpha yeah yeah okay

so um just to take a step back we're

using variational Lelo

approximation so what this technique is

doing is it's taking some distribution

it's going to find the central tendency

of it and then it's going to fit an

inverted Parabola over it and so that

could be a narrow parabola Parabola or it

could be a wide Parabola so that's the

variational Lelo so it is a lot like kind

of fitting a gaussian

approximation but of course a gaussian

has support everywhere and it continues

whereas a parabola like just goes

down um so that's what the llas
approximation
does Theta is being used as commonly to
just refer to the total set of
parameters okay okay now
um down
here this is going to be in the context
of policy
selection so there's going to be beliefs
about action sequences we can take these
beliefs and calculate the average
probability of
pursuing an action
sequence so this is in the spirit of
pursuing action selection as a pure
statistical problem and finding the most
likely path path of least action not
appealing to any kind of like
intermediate reward function or anything
like that this requires us to distribute
the probability for each policy over the
actions implied by that
policy so the array U is going to
connect the actions or the affordances
to the policy
actions or affordances are things that
can happen in a single time step
policies are sequences of multiple time
steps in
length so it's basically like saying
each policy was evaluated so time Step n
and then
um each policy is described in terms of
what actions it contains finally a soft
Max temperature parameter Θ_{sub}
Alpha is applied to account for
Randomness or shaky handedness in
Behavior not accounted for by the

model so this is kind of just brought up here but it's explored more in other papers um when the soft when the softmax is one your behavior is sampled directly from your policy posterior that's matching behavior when the soft Max is very high then you select the best behavior even if it's only slightly better and then if your soft Max is very low it basically erases differences in the posterior likelihood of action okay so so Daniel this this uh it's new parameter is not needed in the um in the active inference model right it's only needed when we um when we do this parameter estimation right when we compute the likelihood function right I mean the core active inference model is very sparse it doesn't contain learning or attention or shaky hand so the textbook is showing us all these different motifs none of them which you have to do but if your phenomena of interest is captured by this type of behavior this is a useful Motif but it's not like any of them are being required for us to do in all situations it's just saying this is one kind of um structural model that can capture decision-making Behavior okay okay but you're you're right like previous examples didn't have this right right I I also would like to

clarify the policy you just mentioned

there so so

policy

um you said the policy specifies

actions uh at each time point if we

think about the discrete time model right

um so it simply says that know what

action I would take at time one what

action I would take at Time

two

um yep here I just went to the ontology

just looked up policy it's a sequence of

actions okay I wonder because in in my

area where I think about policy as a as

a mapping from state to

action uh not simply from time to action

so you're you're thinking about about

you know policies in time one what I do in

time two what I do uh in general uh a

policy defined as in if in certain State

what I should I do time just is one

dimension of the

states um is this a a special feature of

active inference or is it's something

like consistent with the other like

Dynamic uh process

it's more similar than not to how policy

selection is discussed in reinforcement

learning deep Q learning because

fundamentally we're talking about action

selection so it's overwhelmingly more

similar than

not but yes there are some slight

differences because here we're only

talking about policies as sequences of

action through time and then using other

features of the model like other state

um inferences of the model to update our

posterior on

policy but we're we're not we're not

making

policy explicitly a function

of other states other than

time

yeah it's generally the same control

theory setting with just a slightly

different setup but but you you you

should document what is meant by policy

in that setting pull out the quote or

pull out how it's meant and then let's

just just toose it side by side and then

it'll be

clear right in control theory in general

the policies is the action is function

not only the time but also the system

States

right

right

that's right the sequences of actions

are a function of parameters the model

and a sequence of

stimuli that's

here the likelihood of the

parameters is describing the sequence of

actions Through

Time conditioned upon parameters model

and

observations

we can go more into it but it's it's

it's the same setting of inferring

actions from other features of the

model okay so what you what you observes

are this actions

okay

yeah um here here's Us in the behavioral

laboratory observing U Tilda the object

empirical behavior of the system of
Interest objective here it's a objective
meaning
empirical like what we objectively
measured and recorded
empirically in our
laboratory so what we recorded as data
is the output of the subject of our
behavioral
experiment so this isn't taking on the
the the the philosophy of you know
objective and subjective and all this
this is just a behavioral
setup here's the rat that's our
behavioral subject it has a cognitive
Model A generative
model and we're observing Its Behavior
and that is being used as our empirical
scientific
data for cognitive modeling
so θ is also input by the uh
experimental designer by the
um so it's it's given kind of a
parameter right the
stimuli yeah the experimental stimuli is
a action by the
experimenter and a sensory State for the
subject
yeah I think that the the key here that
 θ is the output of the experiment the
behavioral experiment uh and the input
of the the fitting process that gives
you the
 θ once you have done the measurement
 θ you can guess which parameter is the
is the best one to describe this
value right right yeah here is the G
action to find the best parameter θ

right this what the uh or it's the
purpose Theta is the
parameters
um so so the other question I have
actually in this figure 9.1 is what is
M so it seems M just the model all the
parameters of the
model then why why do we use like
different symbols one use M another use
Theta M for the generative model of the
of the subject and Theta for the
experimental
parameters okay for example in the T
example chapter seven what is M and what
is what is Theta what are the
differences M would be the parameters
that describe the cognitive model of the
Rat Theta for the parameters describing
the setup of the task like the idea
that 50% of the time the food is over
here or that when the food is here it
has a 99% probability of
reward okay so M basically represents
other parameters
not uh this is just a schema you know
this is just saying yes this it's you
don't have to use exactly this one and
only Bayesian
graph so it's just saying whatever is
the subjective model you can construct
is you're going to be conditioning upon
the experimental parameters and the
cognitive model
parameters okay experimental um so Theta is
experimental parameters so
those are the parameters that we are
interested in right and they might be
some of the ones you're interested

in
but different Laboratories are
interested in different questions some
might have a very trivial subjective
model and be interested in doing
inference up here other times this might
be fixed up here and there might be
something interesting
here may I interject a question yeah
yeah okay so so Daniel something I I
been thinking about about that's this is
more of a philosophical question
but I really like this
representation uh I just wondered if it
matters in the free energy principle
approach to be worried about
nested cognitive
systems um because in the work that I'm
doing nestedness is uh for the system
that I'm describing there are subsystems
so we have a system within another
system within another
system so from a very abstract point of
view the nestedness is that not relevant
because at every boundary you're just
dealing with a different set of sensory
input from something else so it doesn't
matter if you're messing anything you're
just a system and you can model the
behaviors of that system
separately from the fact that it's
receiving input the inputs are coming
from multiple levels of higher order
that affect it does that make
sense yeah I
mean nested models are are naturally
amenable to this type of modeling okay
so here's a nest an example of a nested

model so here like we we see this kind of figure 4.3 generative model shape um now that that like e downstairs like a downward facing e that's like the sense making component and then the upstairs with a policy selection so now that is kind of expanded so that each Leaf of the E has another nested e and so there's nothing in active inference that says that systems must be nested

however it is very amenable to modeling Collective lateral interactions within a level and then having nested models across levels and then just briefly two kinds of nested models that we see sometimes the nested model refers to an actual spatial enclosure so you know a continent and then a region within a continent and then a village within the region um with like slower and bigger things being at higher levels but also we see nested models for metacognition where you have higher levels of

metacognition as nested observers and and top- Down actors of lower level of cognition so it's not spatially enclosing but it could be thought of as like a cognitive

enclosure I understand that yes that that was very helpful thank you uh but so let me see if I can ask this a little this is very abstract but um is it the case that even so even when we're looking at those nested models um The Nest the sub models the the ones that are nested within a

bigger uh

model there are there any modeling issues that you need to add to to your subsystem in order to take into account that it's receiving sens uh inputs from multiple higher order

systems or does it or you just U from a modeling perspective it doesn't matter you just look at yes they look identical in principle it's well behaved in practice the challenge is that nested models get very very large very quickly

so then it it's like well you know I mean you know the pancreas is nest in the liver and the liver and the chest and the person and the family and it's like all of a sudden you're just with a model with with millions of degrees of freedom with limited data sets so a and yet making a a appropriately articulated nested model actually can help the model embody the structure of the world which uses the data more effectively

so um a smartly constructed nested model will help you make the best of the data you have but a sort of nested model construction as a fishing Expedition ends up losing interpretability even if it ends up recapitulating real

features um yes so no that that's that thank you that was super helpful because actually so one of the boundar Aries to just to give an example one of the boundaries that I'm working with is the boundary between or or thinking about um

the the human body and um the human body
under conditions of our early
evolutionary adaptedness
environment and what's really
interesting about humans is that we have
this

um um brain got access cyc
neuroimmunological sophisticated system
that has a very uh well documented
effect on the

brain uh when uh the immune system is
reacting to a pathogen so it sends
signals to the brain and causes what's
called um stress Behavior right so that's
an example of communication that
we understand that's going on between
subsystems and the

Brain um but we actually can talk about
the communication the way that the
communication is happening so you don't
have to be doing a fishing

Expedition um you you know eventually I
would like to be able to model those
relationships in some way yeah that's a
great transition actually to what um
Andrew um brought up last week so like
since we're in chapter

nine um you know we we're talking about
the empirical data but but the book only
has limited amount of empirical data but
it

cites um heavily these other studies but
then Andrew suggested that we go into
this one so I mean Andrew what what did
you want to discuss about this paper I
pulled out some of the the um figures
that we can talk about but what do you
want to talk about

it sure um see I had a limited amount of time available between last week and this week but the the initial thought was um just kind of because this chapter lays out kind of a series of steps that one could follow as a sort of recipe um the idea was maybe we could um identify in in layman's term well as far as we can in layman's terms like um something like what are what are those steps as played out like for example in this paper here um yeah so we kind of have like we can match like to each step of the recipe provided like maybe what's going on here right awesome yeah so chapter six we constructed the generative model um here and these are all like steps that have to happen but this isn't exactly needing to happen this order but chapter six we basically construct the um generative model that's step two and it's Associated likely Ood function which is just the function of behavior being emitted conditioned upon parameters observations in the generative model like just that we talked about with 9.2 and then we collect data we crunch it all together invert the basian model and then do statistics on the outcome so let's see what it looks like in um the case of Ryan Smith at all's work on gut inference so they develop a asian computational model for gastrointestinal interoception so this is not exactly like the microbiome neurotransmitter slower time scale interaction but this is the faster like vagal nerve

mediated interception with the
gastrointestinal um so it was completed
by healthy individuals they were
undergoing EEG so there was getting
neuro neuroimaging data
and peripheral physiological recording
like C like some measures of the body
and then this I also found this kind of
interesting these self-report questions
how hungry do you feel how you know how
difficult was this task this is just
showing like you can assess a variety of
physiological measures self-report
measures these are all
data okay okay another important thing
that that to keep in mind is like it's
not about constructing the one
generative model it's about having a
portfolio of related generative models
that do and don't have different
components and then you kind of test
across them um like when they talked
about what are the two reasons to do
cognitive modeling one of them was if
you have a fixed generative model then
you can look for group differences
another one is from the data set you can
identify which models are appropriate so
here these 12 models differ kind of
combinatorically in their different
features so standard statistical
modeling technique test for you know
just Mo factor a just Factor B A and B
together A and B and C just C and D
together so so that's what they they're
going to um do and in table one they
Define the same ontology terms that we
see in the

textbook um here's the
test they swallow this
pill that vibrates in their stomach and
then they signal whether they did or
didn't perceive
it so the true
State hidden state
is whether it vibrated or not that's
known because that's controlled by the
experimenter
um and then the observations are the
perception that they
experience obviously if it was a fully
observable situation you'd have like the
a matrix would be an identity Matrix you
would just
simply be able to perceive it but
there's some intensity of signal
where what they're going to test is that
well it turns out that it's not 100%
perceived but also it's not 0% perceived
and maybe there are interesting patterns
of people's
perception here is literally where we
see the basian graph now this is a model
that doesn't
include action because the person is not
in control of vibrating this little
thing so we only are seeing the
downstairs part of the model so in that
way it's
um much like the early example in
chapter
7 so this was the music listening
example in chapter 7 just the
downstairs the prior hidden State
Through Time observations at each time
step but no upstairs action selection

component

prior hidden State on the actual

vibration observations whether it's

perceived a is the mapping between

vibration State and observation B is how

it's transformed so that model could be

specified ahead of time you could

pre-register the study so you kind of

know that you want that and then they

describe this a lot more in the

caption um

they also

describe yeah so then they do some

descriptive statistics on this

model the interceptive

Precision so that's the Precision of the

a matrix if the a matrix is just the

identity matrix it's the most precise

that means that you have perfect

observation if the a matrix was low

Precision it was just a wash you

wouldn't be able to perceive anything

and then it could be anything

between um prior beliefs favoring the

presence of a vibration PV that's the

D

variable and then the learning rate is

describing the the rate of change on

B so those are not simply descriptive

features of the empirical data if you

just wanted to say who's has a higher

accuracy or who's more right or

something you could just summarize the

data you wouldn't even need to do

cognitive

modeling then they look at all of these

different empirical

patterns so here on the columns these

are

um Prima fascia components of the behavioral study these are just descriptive

outcomes from what empirically did happen like the true positive negative rate and the reaction time

patterns on the rows are these cognitive parameters that again they identified with their model so features of their model map not the territory and yet there are still these interesting patterns um then they they can do different

conditions and show that their the cognitive parameters that they're in interested in that those cognitive parameters are variable in different settings here they look at the the cognitively inferred parameters on the rows and then a a uh

self-report and some kind of like um personal data aent

BMI um just like other scientific studies like the kind of result result is like a point estimate and a variance so you can say some of these results are significant at different

thresholds now they're going to connect the cognitive model parameter to EEG

Dynamics here they look at clust string of erps event related potentials so these are kind of like dynamical patterns in the EEG over different time scales and then they can cluster different regions by the similarity of their response and look at how

those kind of traveling waves are
related to
um uh those traveling waves the erps are
related to those different cognitive
modeling
parameters
and they do that for other brain regions
too me then D can I ask a question
yeah the the um correlation Matrix showed
uh sorry the before with the learning
rate is λ rate estimated from the uh
experimental
data yes everything is estimated from
the data in this
paper okay
okay so so how how do they construct the
magn function to do the estimation we
use the um EG data to to
estimate I mean you look look at the
paper for
more details but
um the EEG data are being taken as kind
of
like
this is you know what you see is what
you get here right there's no EEG here
there's just a prior on vibration and
then an observation and then the
mapping between the observation and the
vibration or not so that's kind of like
the core phenomena that they're studying
and then they have the age and the BMI
and the EEG and the
self-report so then those other data are
kind of
overlaid on the timeline of this kind of
core model and then you can look for
correlations

[Music]

between

um this graph and the

EEG so I don't think you need to take in

the likelihood of the EEG data

specifically well but in order to

estimate at the learning

rate they do need a MAG function right

so it has just parameter of this activ

inference

model yeah I mean we can look for for

more details on how the learning rate is

calculated but they describe it in that

caption I mean

it's it's described but yeah look into

it and see if that's how you would

describe learning rate and these are

features of the map not the

territory so again it's common for

people to look at this and then

reference some feature of the

world but it's not about referencing

some feature of the world to validate or

invalidate this model it's about

generating a portfolio of models that

have different

features and then propagating that

portfolio of models

forwards and then saying actually

learning rate does matter because the

one that had it versus the one that

didn't had this better

fit and someone says oh but you didn't

think about this and you go okay cool

now let's add model 13 with that

feature and that's the real that's the

Delight of the empirical

conversation just people bring things up

and it's like okay either we have explored it and we can talk about the impact of that variable or we haven't considered it yet so then thank you for the suggestion let's add it in and then come back to it and that's the iterated modeling process but yeah here by looking at the table how how do we how do we know which model is a better description um so you have 12 models to yeah yeah yeah great question they they describe this the most common method used is the basian information Criterion Bic or the related um AIC um the Bic criteria basically um rewards fitting a model uh fitting the data well and it penalizes having extra parameters and so because you're all every model with another parameter you will always get a better fit like a principal component analysis the next principal component all always is going to soak up more variants so make the best fitting model is actually a kind of

[Music]

um ironically useless singular goal but if you say I want the model to be at the optimal tradeoff where like the parameters are like punching above their weight so that's usually how people select it but that's not the only model selection imperative you might say I'm okay with having a huge model I just want the most accurate one so then you might take one that has more parameters

or you might say I don't need the most accurate one I need it to just be better than

50% and I need it to you know fit on an index

card yes there isn't an answer in the general case for which model you should select and you might want to actually just hold the whole portfolio and just keep that whole distribution that's like Ensemble

modeling yeah so so I'm just wondering why it doesn't show the selection criteria in the table when they selection

criteria yeah yeah so the BC they didn't show us right um well not in that table like I I mean I the the models are compared in some formal way but in that image they're just they're just listing which ones they are but yeah it could have included like the likelihood of the Bic for the models or the number of parameters

um it's just not shown in that representation maybe in the supplement or maybe somewhere else but but yeah that that's usually what is shown you see like it will be like model one and it will be like Delta Bic Z and then it will show how how much behind in the Bic the other ones

are um and then if we were studying an active setting like let's just say that now we're in a little modified version where the person can make can be in the you know red or the orange room and they can move between those two

rooms and then those two rooms have like a different probability of the stomach thing vibrating so now there's some policy variable well that's the power of the base graph it's just another parameter to estimate so the same ex and then someone says well what about having a policy on the air conditioning as well then you add the air conditioning policy oh and now we want the observations not just to be the interception now we want to have like you know extra reception too you just keep composing and building the models but in chapter nine that's where we make the contact with the empirical and that kind of like reminder that there are real things to study and that you can design the generative model ahead of time and then do the experiment and then just as soon as you collect the data you just dump it into the generative model and you basically have your paper ready that's one great option another option is the data have already been collected and you know what structure they are and then you can construct a generative model that will load in that structure so whether the data are collected before or after obviously it's different scientifically but in both cases this chapter gets us to the point where we can take in data bring it into

this kind of
format but then in the end we make the
kind of bar bar plots and time series
and correlation matrices that we would
see like from literally any other
scientific paper
yeah what what did anyone think about
this well I'm just kind of stepping back
and I wanted to uh ask
you uh so is a lot of the literature
that
on recent literature on the energy
principle about you know when we going
to the empirical world is figuring
out or trying to infer what generative
models different
systems have in order for us to be able
to begin to test hypothesis from the
free energy
prot
that's a great question yeah definitely
like one line of research or multiple
lines of research are related to the
fundamental physics of cognition some
lines of research are more computational
capacity
building and so they're interested in
like motifs and comp computational
implementations another set of research
that's more empirically oriented is
looking at empirical data and developing
cognitive
models that's one direction of
empiricism and then the other direction
of empiricism is with the cognitive
model in hand looking at differences
amongst individuals or
groups and like Ryan Smith ated all's

work has both of those directions of empiricism sometimes um like in this paper they're taking a simple setting and then they're exploring a range of cognitive models in other settings they're basically taking a cognitive model identified from previous work or however and then asking okay do individuals with these different diagnoses differ for example and those those two directions of empiricism can coexist in one paper but but then it's kind of like this joint identification of what should the model be but then how do the individuals differ in the model and here's those two reasons that they say computational phenotyping with the model in hand describing individuals or with the data in hand ask which model you want to utilize Daniel can can I ask you a question here yeah about T's example um in chapter seven the textbook provides a complete example about the uh active influence model and in this chapter um they also talk about that t example but I wonder if there is any empirical data about the the mouse you know movement and that can we can like you know use the same example to illustrate the the basic steps and idea in chapter

nine yeah I mean it's a great question something I've wondered about too don't we have like hundreds of thousands of data points on the teamas I don't know what data set that is but but these are experiments that have been done here those are simulated data not the experimented data right yeah here they just were using the teamas in the synthetic Direction but yeah it's a great project to find an empirical data set um and then you would need to make the generative model for it because it wouldn't be exactly this but this structure might work well but yeah I mean this is the work that I hope that the textbook gets us all to is going to behavioral settings and studies that have been done and building generative model but this is literally the work that we have to do well you're you're not aware of anyone have done it right because I I asked a few people who who uh extend this uh teamas uh example and but none of them have done empirical um study and estimate The Motto parameters I've never seen it with like the mammal Tas again even though they're they're are actually probably thousands of data sets on the Tas so I think that's a a very viable option perhaps people use the teamas

because it's easy to make a point about epistemic value but usually the teamas isn't the case that people are interested in studying but if it were a lab that were studying decision making or addiction or learning in rodents then they would be actually looking to do a cognitive model with the teamas so that's where the collaboration between people with active inference expertise and domain experts and then you have an iterated modeling process with

[Music]

them are all the features included in this model and the back and forth there and then find out what the generative model what kind of empirical insights you can have and in that case if it's already a data set that's collected in a published paper that's the perfect setting because then you can say here's what active inference goes beyond descriptive statistics and what it actually what value it does add to the setting yeah yeah there's a lot of Behavioral data sets to do this on yeah if you were actually looking for data sets I mean there's a great um the busak labs there's um Yori rousaki neuroscientist at NYU he has an entire lab dedicated to um things like for example studying like mice and and rats are kind of his

expertise and um you know running many experiments where they're trying to use um electrophysiological data to track uh what is going on in mice what's going on in their their neurons whenever they're navigating a maze and it's where they do research on things like um Place cells and all these fascinating things where if you draw out a topological map out of the EEG data you can actually see like the the the mouse is tracking where it is location

so I see Daniel's point to bring it back that there are many experiments done that highly relate to the tas's example but I guess it's around there that you could have collaboration between someone who's an expert in active inference versus those who carry out these kinds of experiments to test the data that's already been collected uh but within more of active inference kind of framing because because the the the lab technician here at busak or otherwise are attempting to understand something more like the neurobiology but they're not necessarily getting too indepth with the cognitive modeling aspect or the kinds of questions that someone um doing research in active inference might be asking that's my thought on that right yeah thanks

angel

so

um this is a uh this is a question that is bugging me and I just want to bring it up now um before I get really deep into all the material which is um when

working with human

subjects uh you know one of the things

that that that I see as a is it's always

been a huge problem in out my research

is that as soon as you enter into the

interview

interview

setting that in itself you know it has a

huge observer effect so so from the

perspective of active inference you're

dealing with the fact that this

individual from whom you're getting some

information about their generative model

in their heads they have a generative

model in their heads about that

interaction and making and already

trying to make assumptions or thinking

about what is this interviewer going to

ask me and what are their expectations

because that's what we humans do and so

if you could tell me early on where in

this world of active influence is that

addressed because it's a huge issue in

in my systems in the systems that I work

with

yeah it's a big question there's a lot

lot of pieces to say on it okay um

suffice to say that studying like a

pendulum is an easier active system

than study an active system that has

theory of

mindly okay thank you thank you that was

like really well but so yeah it's not to

be

found in this image but this image is

kind of like yeah pointing towards it

okay yeah thank you to the textbook

group

um we return for chapter 10 next time

okay thank you very much take care bye

bye